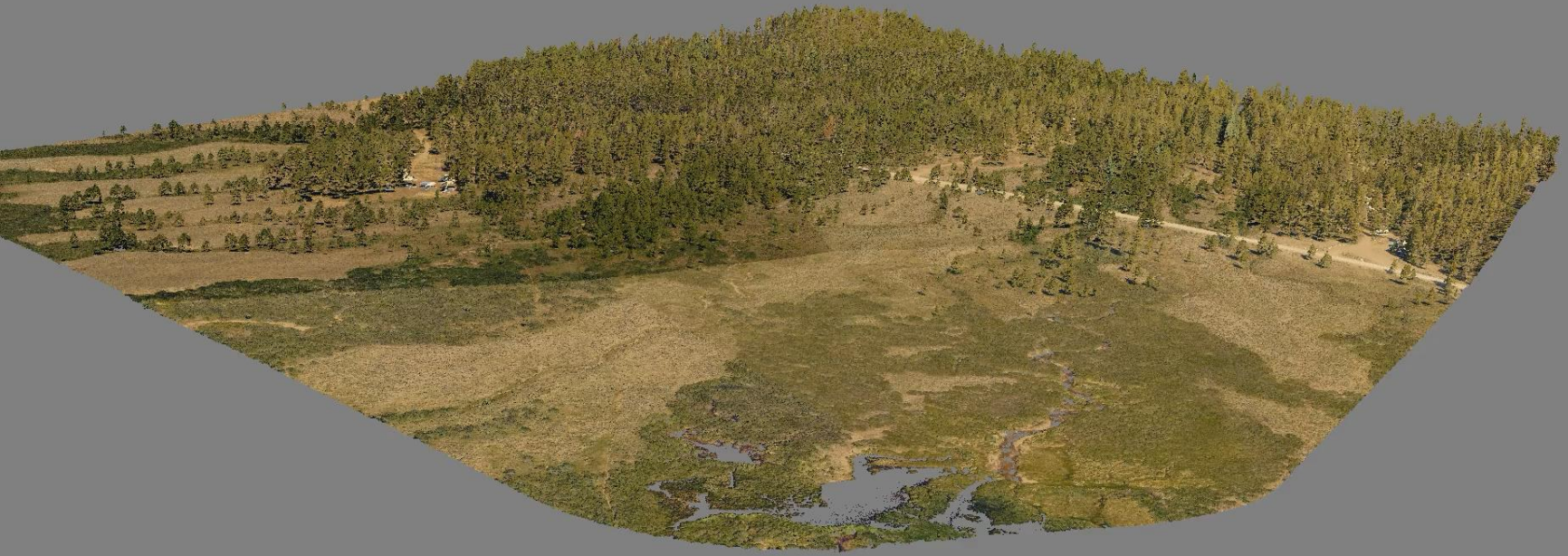
CHESS Workshop 2026
Intro to LiDAR

March 12, 2025

BERKELEY LAB · SLAC · Berkeley UNIVERSITY OF CALIFORNIA · DRI Desert Research Institute · MIT Massachusetts Institute of Technology · Oregon State University · Stanford · Rocky Mountain Biological Laboratory · U.S. DEPARTMENT of ENERGY



What's LiDAR again?

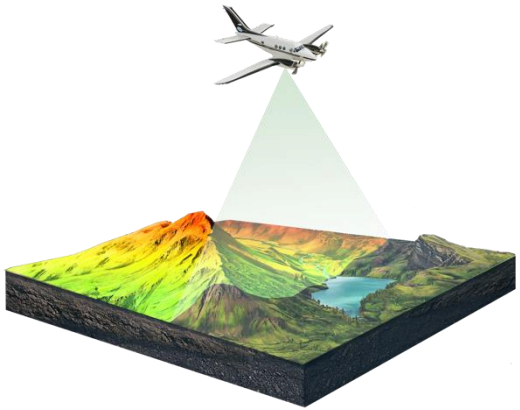
Light **D**etection **a**nd **R**anging

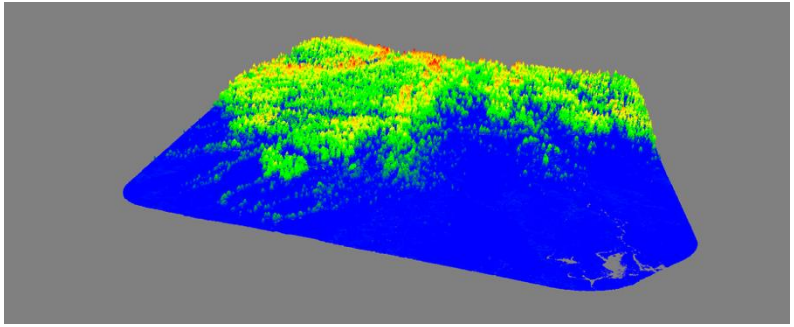
1. Remote sensing method used to examine Earth's surface
2. Direct measure of the target surface profile in three dimensions
3. Active sensor: provides its own energy source

What's LiDAR again?

Top-down

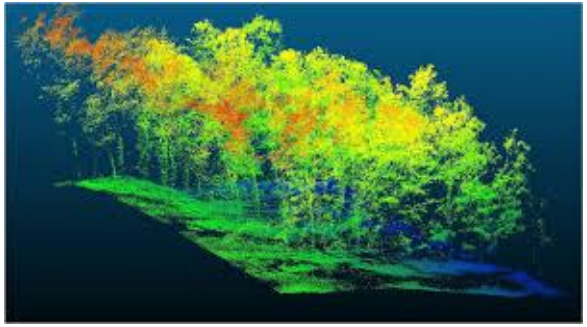
Airborne Laser Scanning [ALS]



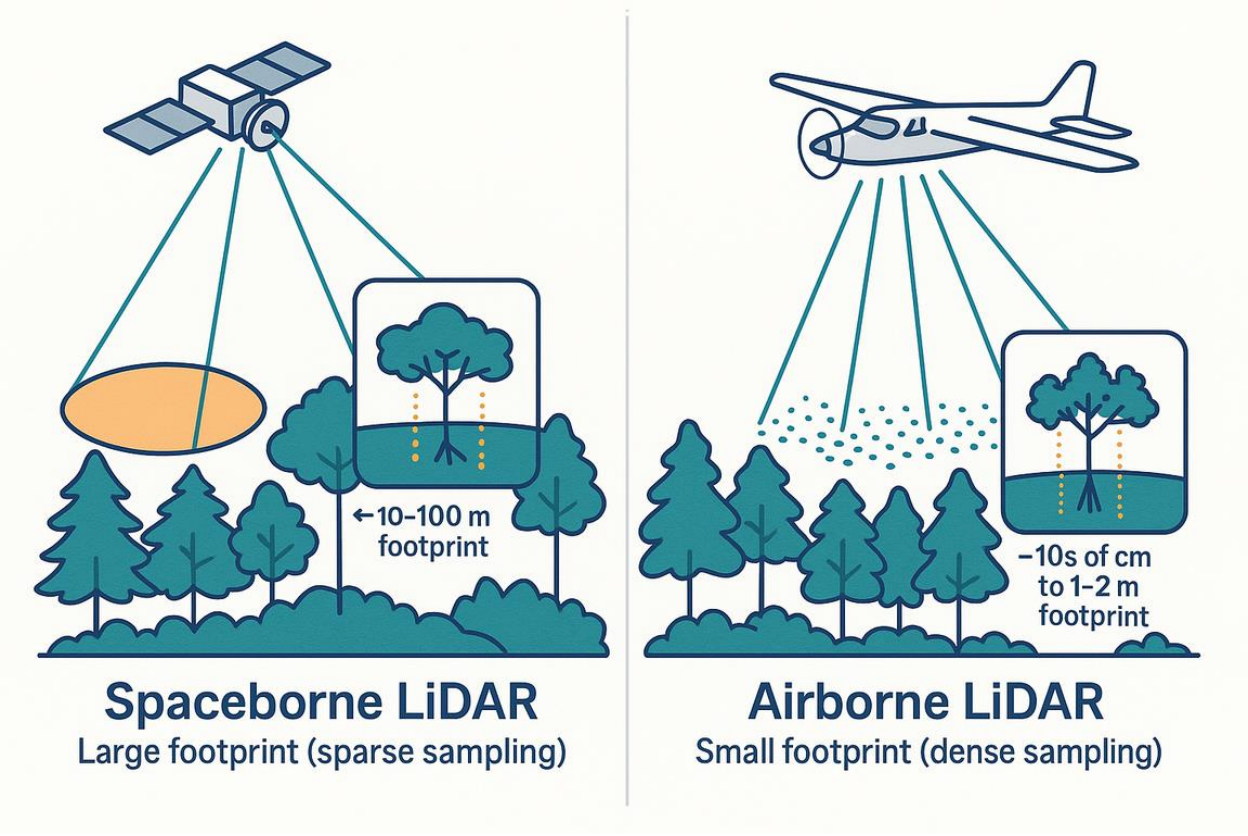


Ground-based

Terrestrial Laser Scanning [TLS]



What's LiDAR again?



NEON Airborne Observation Platform (AOP)

Optech Galaxy Prime

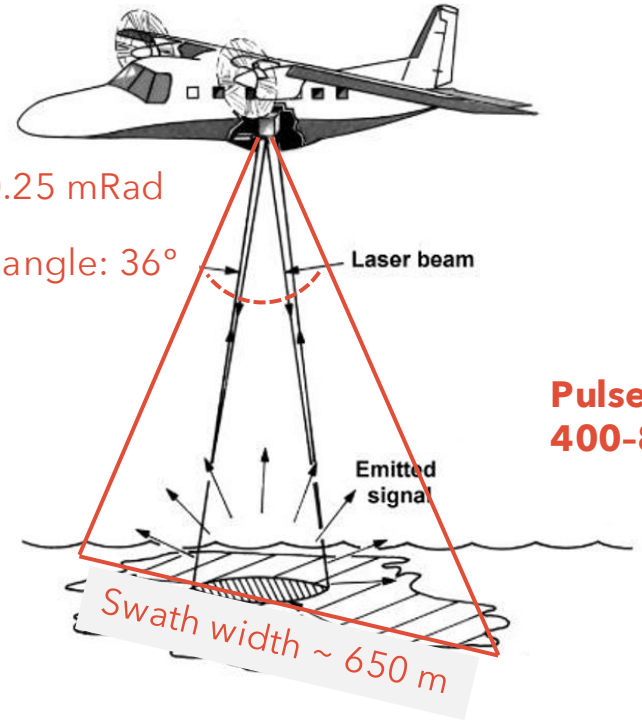
NEON Airborne Observation Platform (AOP)

Altitude: 3600 – 4600 m
(~ 1000 m above ground)

Wavelength: 1064 nm
(near infrared)

Beam divergence: 0.25 mRad

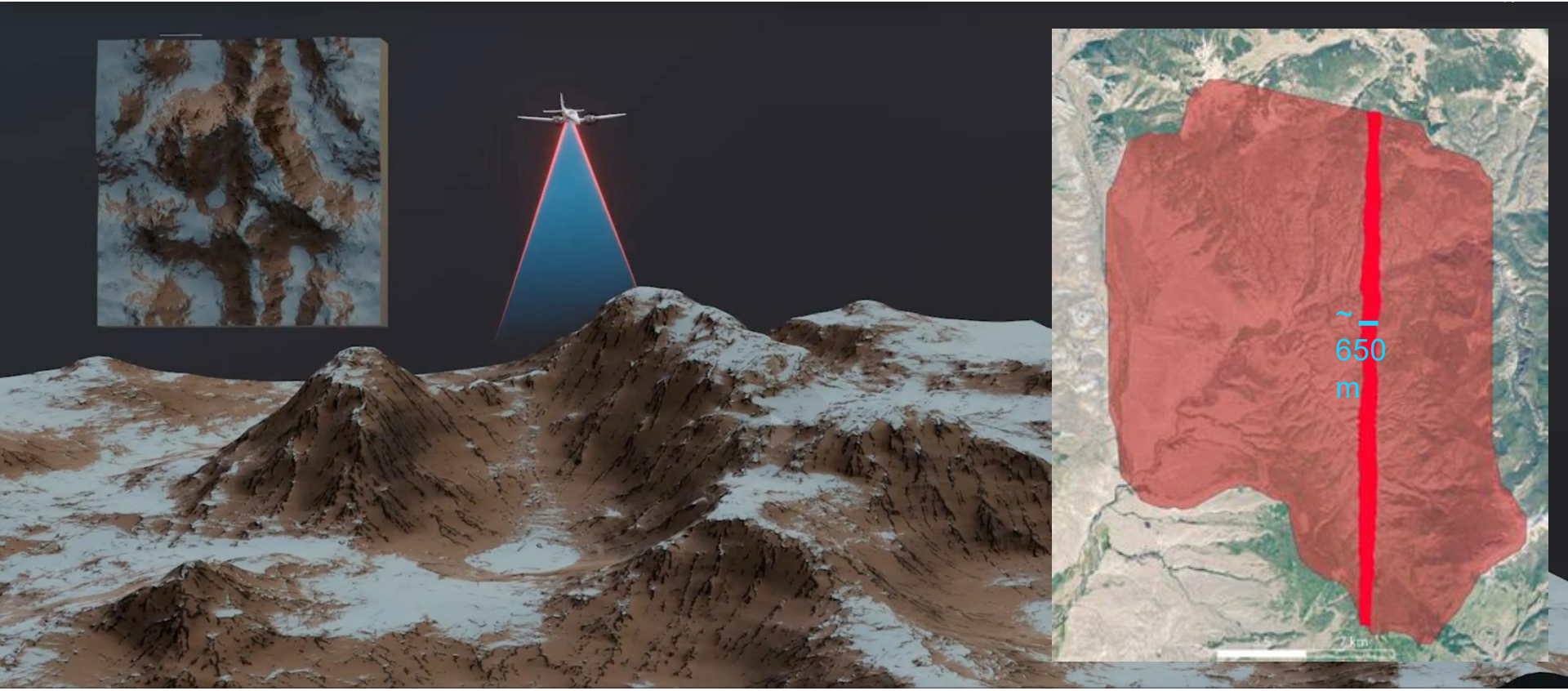
Scan angle: 36°



Pulse repetition frequency (PRF)
400-850 kHz

Ground footprint diameter ~ 0.25 m to 0.8 m

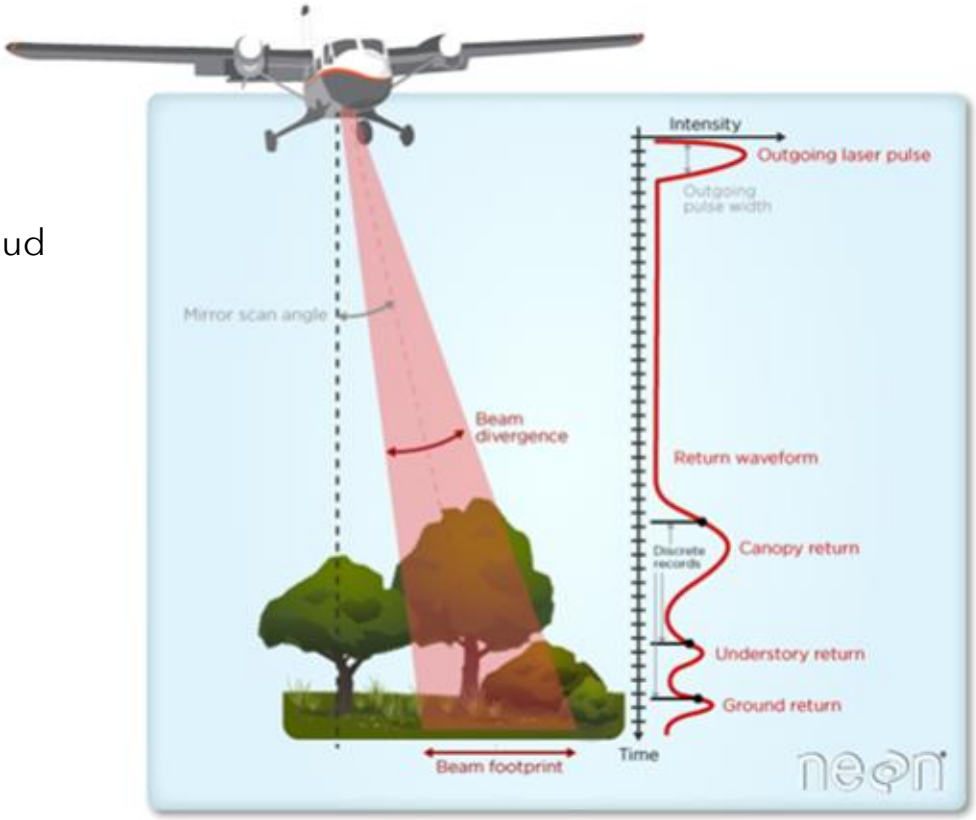
NEON Airborne Observation Platform (AOP)



Data modalities

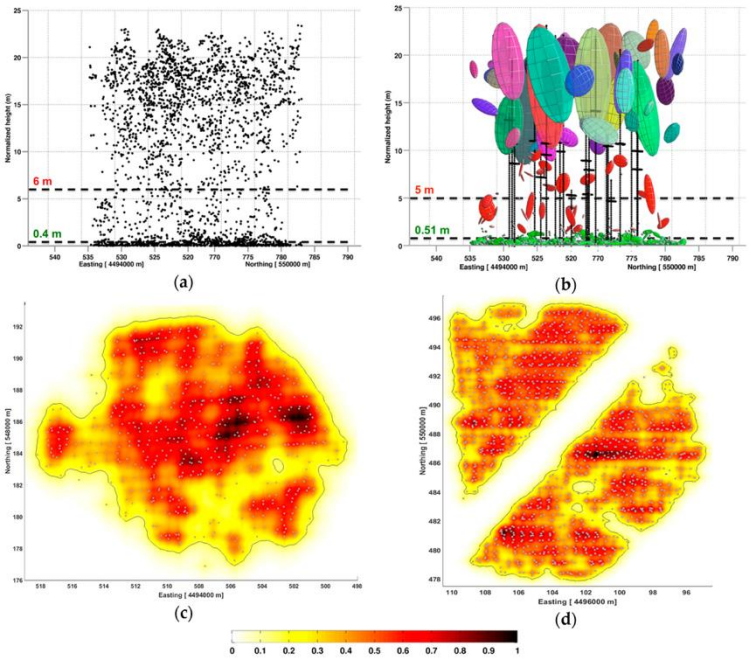
Discrete point cloud

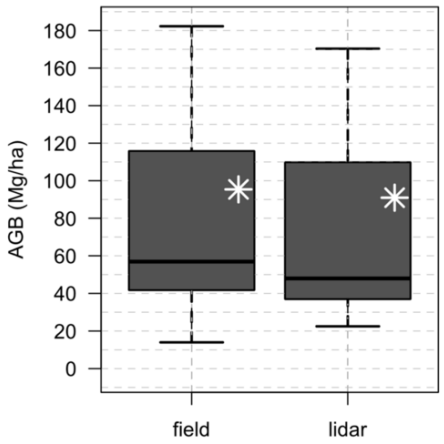
Full waveform



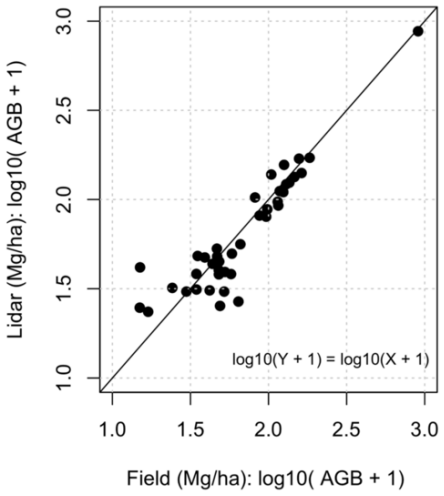
What can we do with LiDAR data?

Estimate aboveground biomass in the absence of field inventory data





(a)

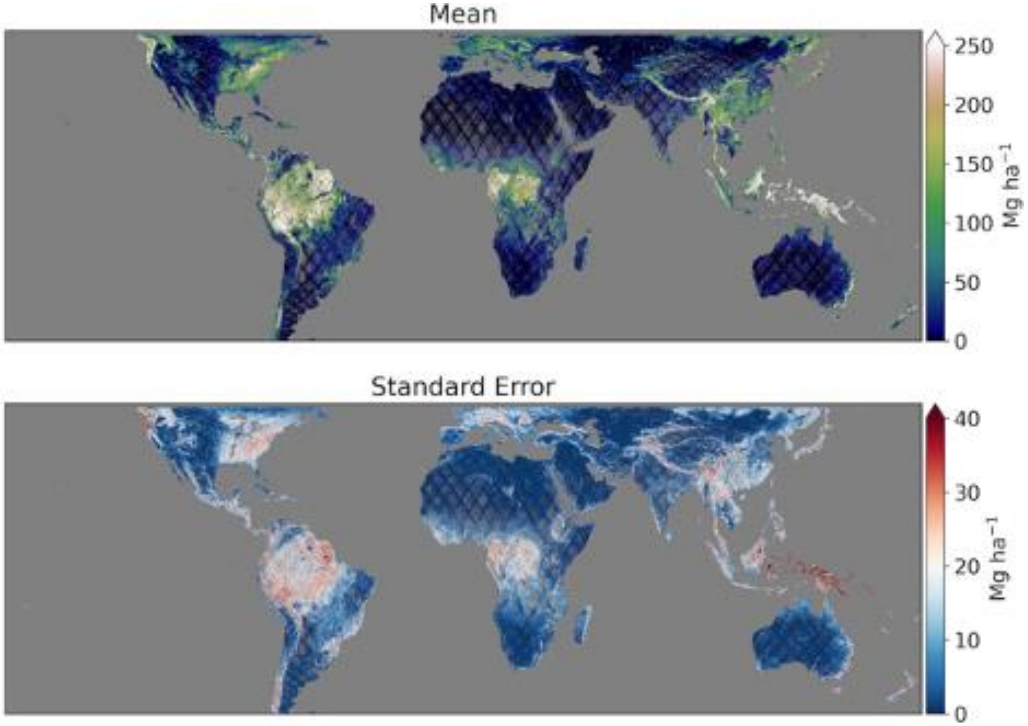


(b)

Ferraz et al. 2016. *Remote Sensing*

What can we do with LiDAR data?

Estimate biomass at global scale



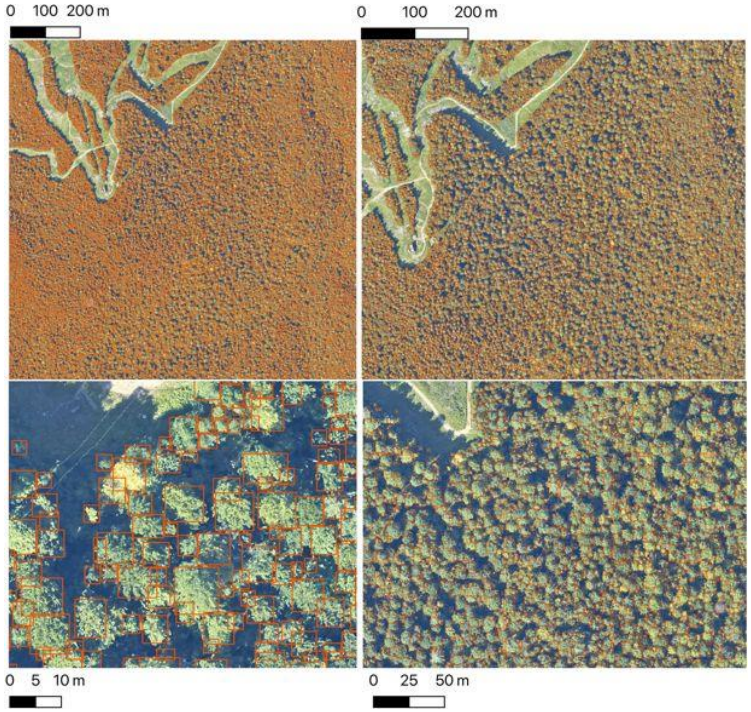
NASA 2021

NASA GEDI Biomass Map

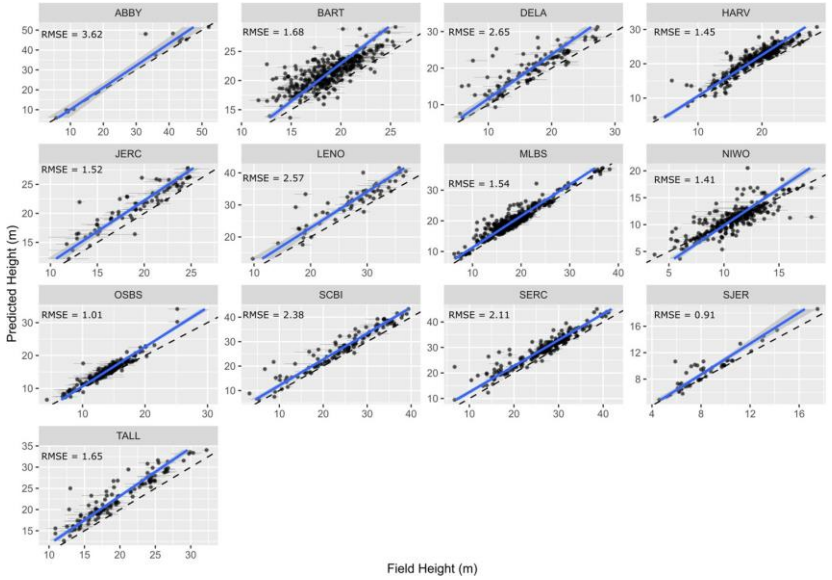
A 1x1 km spatial resolution mean aboveground biomass density (AGBD) map derived from GEDI lidar data

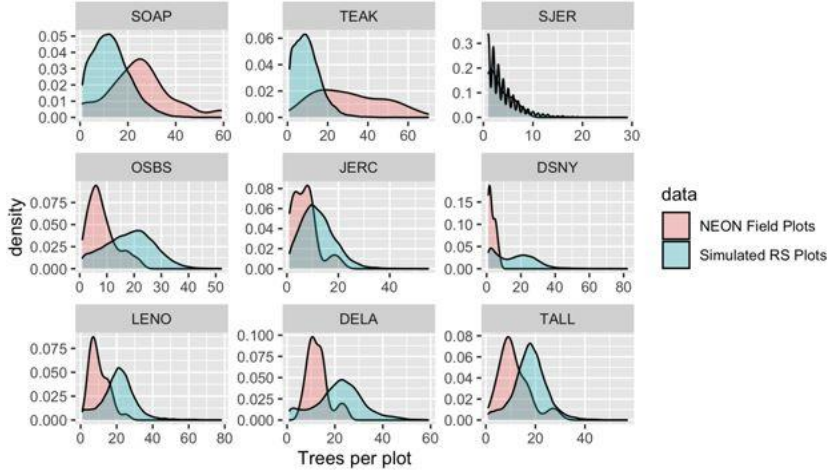
What can we do with LiDAR data?

Map all the trees



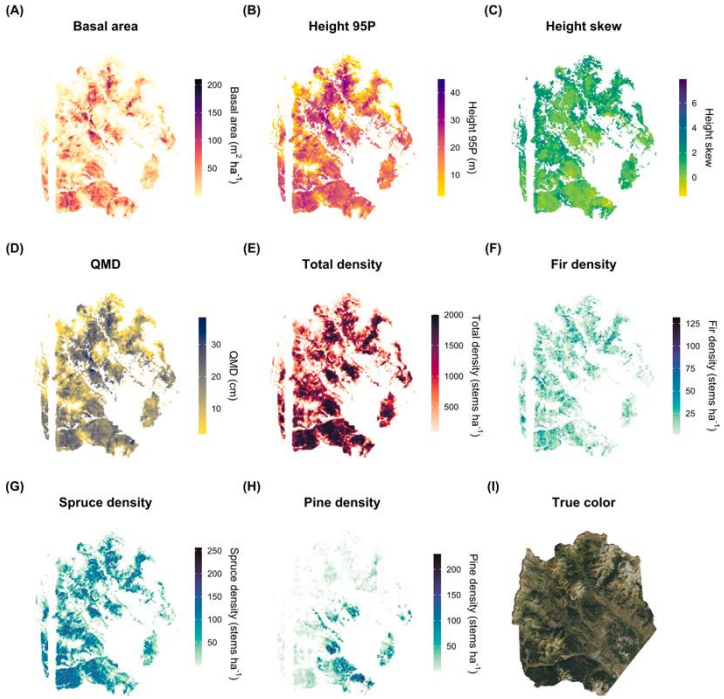
Weinstein et al. 2021. eLife

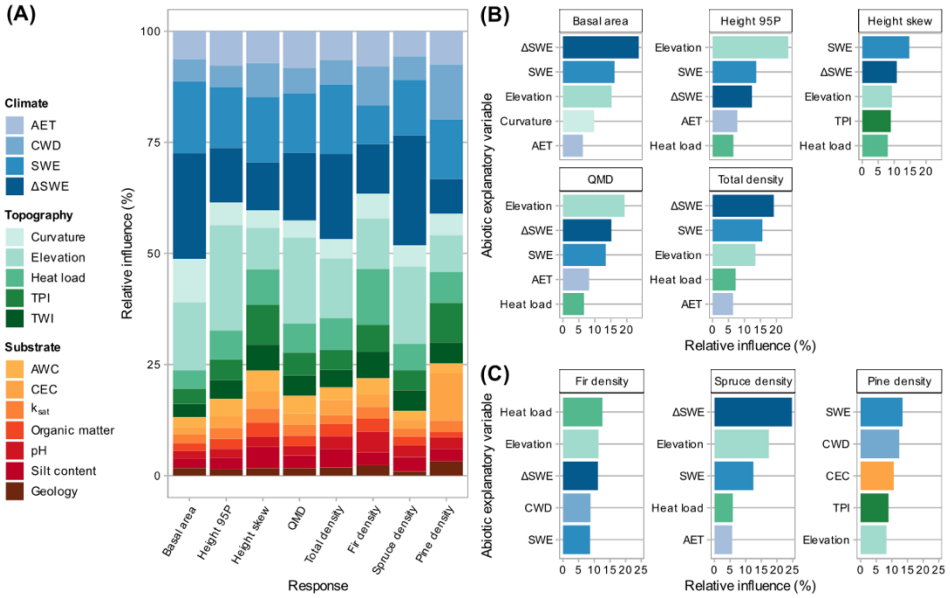




What can we do with LiDAR data?

Map forest structure and understand its physiographic drivers

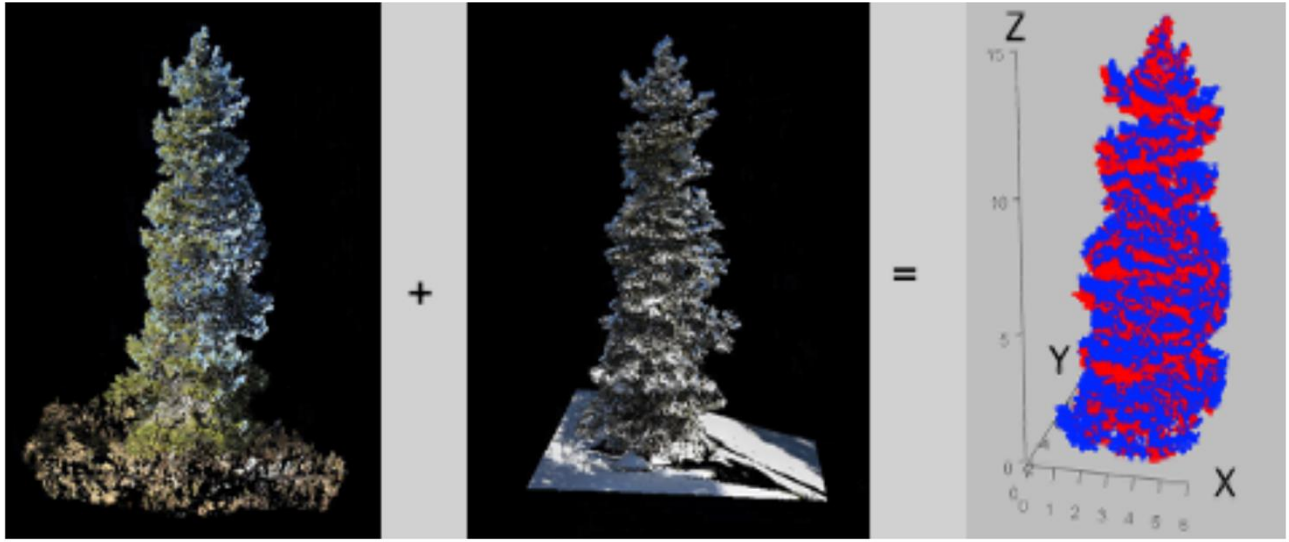


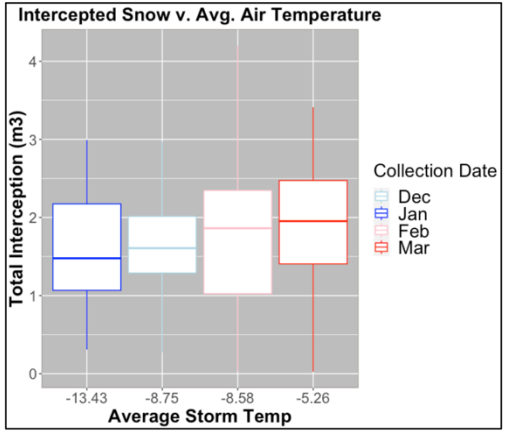


Worsham et al. 2025. RSE

What can we do with LiDAR data?

Quantify snow interception

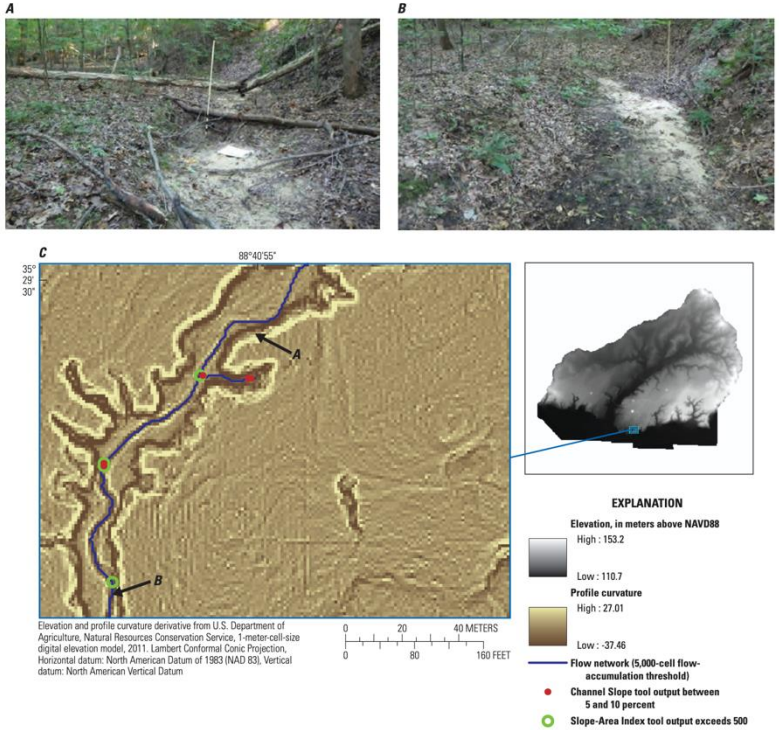




Beutler 2024. Master's Thesis

What can we do with LiDAR data?

Describe fine-scale stream channel hydrogeomorphology



Cartwright and Diehl. 2016

Data structures

LAS/LAZ v 1.4

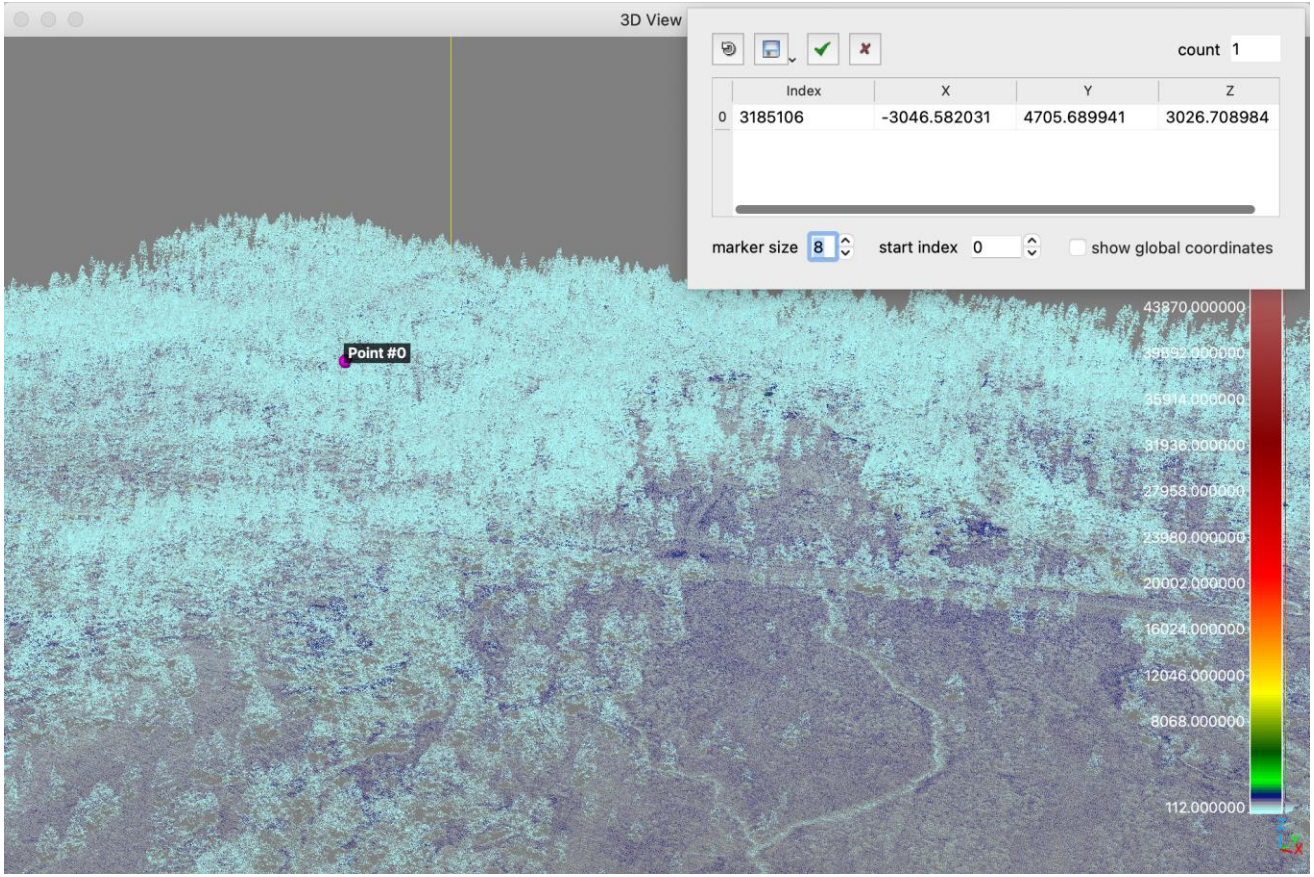
header
data payload
crs
index

```
Public Header Block
File signature:          LASF
File source ID:          0
Global encoding:
 - GPS Time Type: GPS Week Time
 - Synthetic Return Numbers: no
 - Well Know Text: CRS is WKT
 - Aggregate Model: false
Project ID - GUID:       00000000-0000-0000-0000-000000000000
Version:                 1.4
System identifier:       LAStools (c) by rapidlasso GmbH
Generating software:     lascolor (210418) commercial
File creation d/y:       205/2025
header size:             375
Offset to point data:    1288
Num. var. length record: 3
Point data format:       7
Point data record length: 36
Num. of point records:   40902978
Num. of points by return: 39787116 1113342 2519 1 0 0 0 0 0 0 0 0 0 0
Scale factor X Y Z:      0.001 0.001 0.001
Offset X Y Z:            330000 4300000 0
min X Y Z:               327000 4315000 2910.752
max X Y Z:               328000 4316000 4557.23

Variable Length Records (VLR):
  Variable Length Record 1 of 3
    Description:
    WKT OGC COORDINATE SYSTEM: PROJCS["WGS 84 / UTM zone 13N",GEOGCS["WGS
84",DATUM["wgs_1984",SPHERO [...] (truncated)
  Variable Length Record 2 of 3
    Description: PROCESSING
  Variable Length Record 3 of 3
    Description: PREVIOUS GENERATING SOFTWARE
Extended Variable Length Records (EVLR): void
```

Data structures

LAS/LAZ v 1.4



Data structures

LAS/LAZ v 1.4

header
data payload
crs
index

X	double [40902978]	327000 327001 327001 327000 327000 327000 ...
Y	double [40902978]	4315001 4315001 4315001 4315001 4315001 4315001 ...
Z	double [40902978]	2932 2932 2932 2932 2932 2932 ...
gpstime	double [40902978]	405972 405972 405972 405972 405972 405972 ...
Intensity	integer [40902978]	736 784 704 848 816 816 ...
ReturnNumber	integer [40902978]	1 1 1 1 1 1 ...
NumberOfReturns	integer [40902978]	1 1 1 1 1 1 ...
ScanDirectionFlag	integer [40902978]	1 1 0 0 0 1 ...
EdgeOfFlightline	integer [40902978]	0 0 0 0 0 0 ...
Classification	integer [40902978]	2 2 2 2 2 2 ...
ScannerChannel	integer [40902978]	0 0 0 0 0 0 ...
Synthetic_flag	logical [40902978]	FALSE FALSE FALSE FALSE FALSE FALSE ...
Keypoint_flag	logical [40902978]	FALSE FALSE FALSE FALSE FALSE FALSE ...
Withheld_flag	logical [40902978]	FALSE FALSE FALSE FALSE FALSE FALSE ...
Overlap_flag	logical [40902978]	FALSE FALSE FALSE FALSE FALSE FALSE ...
ScanAngle	double [40902978]	-13.6 -13.6 -13.3 -13.3 -13.3 -13.6 ...
UserData	integer [40902978]	0 0 0 0 0 0 ...
PointSourceID	integer [40902978]	33 33 33 33 33 33 ...
R	integer [40902978]	25856 38144 39680 51456 31744 36096 ...
G	integer [40902978]	26368 38912 39168 50688 32256 35584 ...
B	integer [40902978]	13312 28416 27648 36096 18944 22784 ...

Data structures

LAS/LAZ v 1.4

header
data payload
crs
index

```
Coordinate Reference System:
  User input: PROJCS["WGS 84 / UTM zone 13N",GEOGCS["WGS 84",DATUM["wgs_1984",SPHEROID["WGS
84",6378137,298.257223563,AUTHORITY["EPSG","7030"]],TOWGS84[0,0,0,0,0,0,0],AUTHORITY["EPSG","6326"]],PRIME
M["Greenwich",0,AUTHORITY["EPSG","8901"]],UNIT["degree",0.0174532925199433,AUTHORITY["EPSG","9102"]],AXIS[
"Geodetic longitude",EAST],AXIS["Geodetic
latitude",NORTH],AUTHORITY["EPSG","4326"]],PROJECTION["Transverse_Mercator"],PARAMETER["false_easting",5000
00],PARAMETER["false_northing",0],PARAMETER["central_meridian",-
105],PARAMETER["scale_factor",0.9996],PARAMETER["latitude_of_origin",0],UNIT["Meter",1,AUTHORITY["EPSG","900
1"]],AXIS["Easting",EAST],AXIS["Northing",NORTH],AUTHORITY["EPSG","32613"]]
  wkt:
BOUNDCRS[
    SOURCECRS[
        PROJCRS["WGS 84 / UTM zone 13N",
            BASEGEOGCRS["WGS 84",
                DATUM["World Geodetic System 1984",
                    ELLIPSOID["WGS 84",6378137,298.257223563,
                        LENGTHUNIT["metre",1]]],
                PRIMEM["Greenwich",0,
                    ANGLEUNIT["degree",0.0174532925199433]]],

                            ...

                    ID["EPSG",8801]]],
            CS[Cartesian,2],
                AXIS["easting",east,
                    ORDER[1],
                    LENGTHUNIT["Meter",1]],
                AXIS["northing",north,
                    ORDER[2],
                    LENGTHUNIT["Meter",1]],
            ID["EPSG",32613]]],
```

Data structures

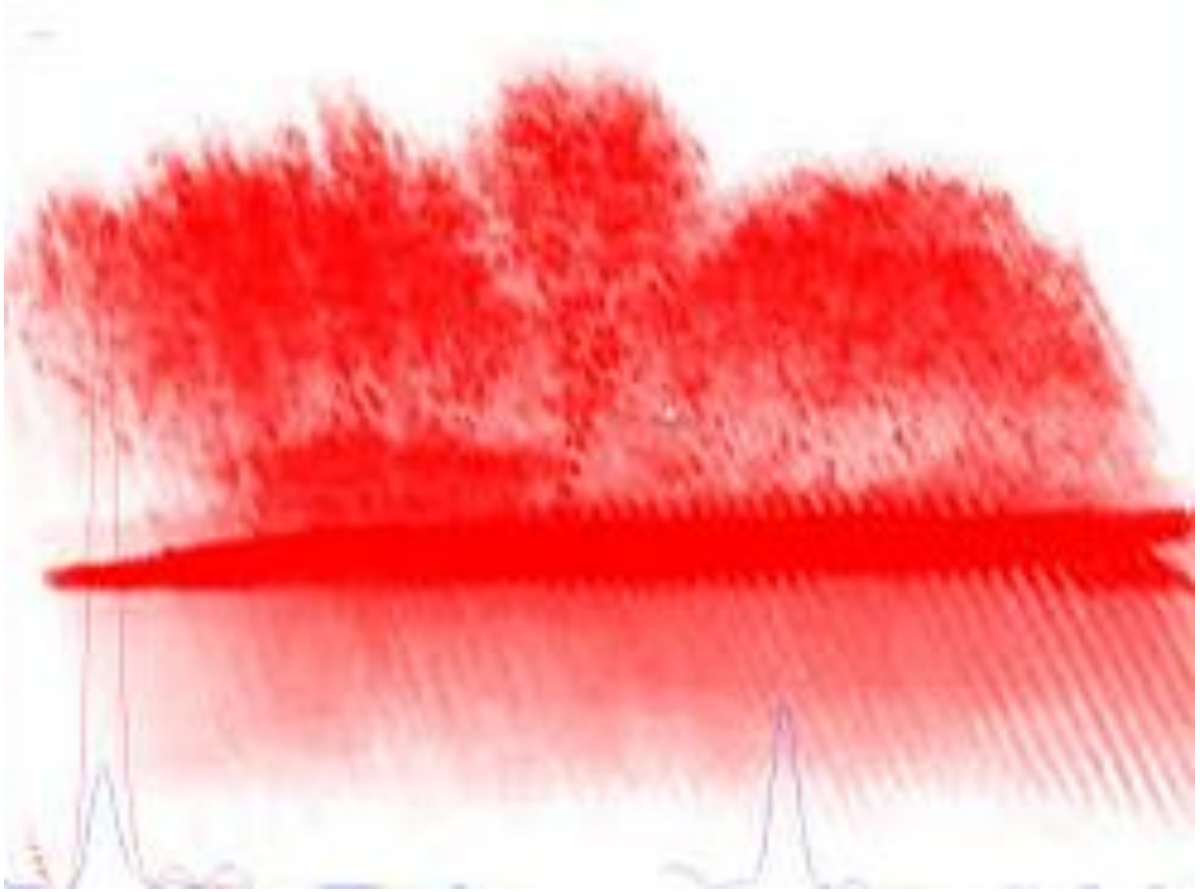
LAS/LAZ v 1.4

header
data payload
crs
index

EMPTY

Data structures

ENVI Binary Waveform



Data structures

ENVI Binary Waveform

13 file pairs
- header
- binary data payload

2013_SJER_AOP_FL03_waveform_ephemeris_array_img
2013_SJER_AOP_FL03_waveform_ephemeris_array_img.hdr
2013_SJER_AOP_FL03_waveform_first_return.las
2013_SJER_AOP_FL03_waveform_geolocation_array_img
2013_SJER_AOP_FL03_waveform_geolocation_array_img.hdr
2013_SJER_AOP_FL03_waveform_observation_array_img
2013_SJER_AOP_FL03_waveform_observation_array_img.hdr
2013_SJER_AOP_FL03_waveform_outgoing_pulse_array_img
2013_SJER_AOP_FL03_waveform_outgoing_pulse_array_img.hdr
2013_SJER_AOP_FL03_waveform_return_pulse_array_img
2013_SJER_AOP_FL03_waveform_return_pulse_array_img.hdr
2013_SJER_AOP_FL12_waveform_ephemeris_array_img
2013_SJER_AOP_FL12_waveform_ephemeris_array_img.hdr
2013_SJER_AOP_FL12_waveform_first_return.las
2013_SJER_AOP_FL12_waveform_geolocation_array_img
2013_SJER_AOP_FL12_waveform_geolocation_array_img.hdr
2013_SJER_AOP_FL12_waveform_observation_array_img
2013_SJER_AOP_FL12_waveform_observation_array_img.hdr
2013_SJER_AOP_FL12_waveform_outgoing_pulse_array_img
2013_SJER_AOP_FL12_waveform_outgoing_pulse_array_img.hdr
2013_SJER_AOP_FL12_waveform_return_pulse_array_img
2013_SJER_AOP_FL12_waveform_return_pulse_array_img.hdr
2013_SJER_FL03_waveform_impulse_response_array_img
2013_SJER_FL03_waveform_impulse_response_array_img.hdr
2013_SJER_FL03_waveform_impulse_response_T0_array_img
2013_SJER_FL03_waveform_impulse_response_T0_array_img.hdr

Data structures

ENVI Binary Waveform

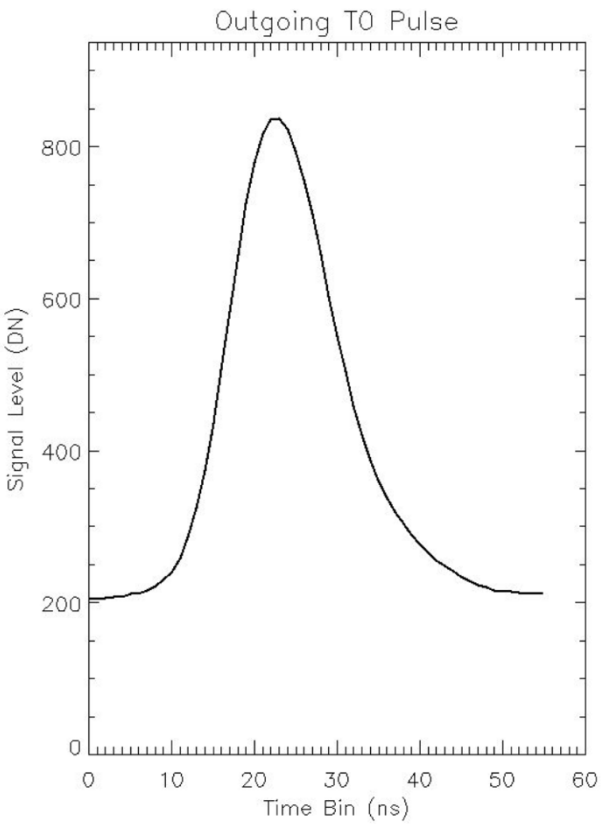
13 file pairs
- **header**
- binary data payload

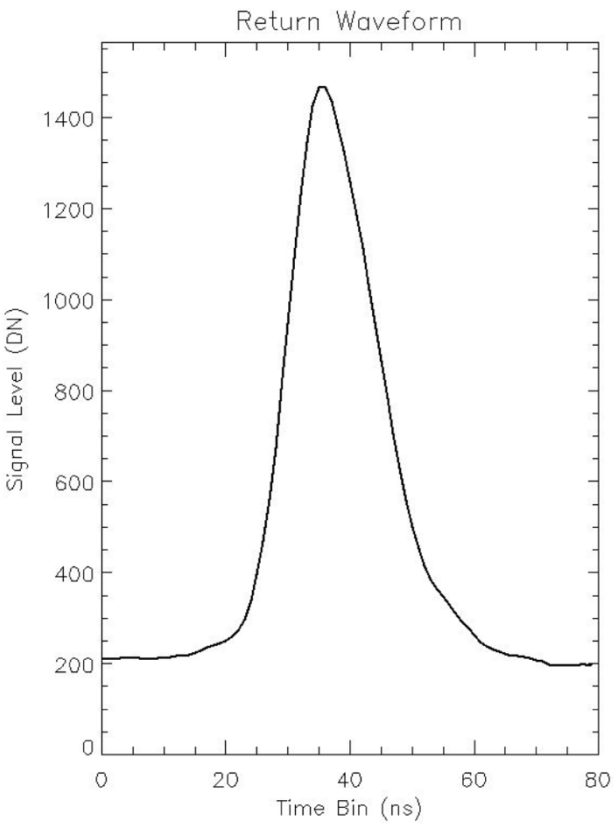
```
ENVI
description = {
NEON Waveform LiDAR Outgoing Pulse File.
[Tue Jan 27 13:52:22
2015] }
samples = 100
lines = 167019
bands = 1
header offset = 0
file type = ENVI standard
data type = 2
interleave = bil
sensor type = Unknown
byte order = 0
wavelength units = Unknown
band names = { Outgoing Pulse }
```

Data structures

ENVI Binary Waveform

13 file pairs
- header
- **binary data payload**





Tutorial

https://github.com/rmbl-chess/chess_workshop

1. Clone the repo or download demos/lidar_demo
2. Open chess_lidar_tutorial.Rmd in Rstudio
3. Set Tools > Global Options > R Markdown > Evaluate chunks in directory: Document